

MERN Stack Developer Interview Questions (2+ Years Experience)

General MERN Stack Questions

- What is the MERN stack, and why is it popular?
- How does the MERN stack differ from the MEAN stack?
- Explain the role of each technology in the MERN stack.
- How do you manage full-stack architecture in your projects?

React.js (Frontend)

- What is the virtual DOM and how does React use it?
- What are hooks in React? Explain useEffect and useState.
- How do you manage global state in React?
- Difference between functional and class components?
- How do you optimize performance in a large React app?
- Explain controlled vs uncontrolled components.
- How do you handle form validation?
- What are React keys and why are they important in lists?

Node.js and Express.js (Backend)

- How do you create RESTful APIs using Express.js?
- What is middleware in Express? Give examples.
- How do you handle file uploads in Node.js?
- How do you manage error handling in Express?
- How do you secure a Node.js app (e.g., helmet, rate limiter)?
- Explain the concept of asynchronous programming and how it's handled in Node.
- How do you implement JWT authentication?

MongoDB (Database)

- Difference between SQL and NoSQL databases?
- How do you define schema using Mongoose?
- How do you handle relationships in MongoDB (ref vs embedded)?
- How do you perform aggregation in MongoDB?

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- What indexing strategies have you used?
- How do you ensure data validation in MongoDB?

DevOps & Tools

- How do you deploy a MERN app (e.g., on Heroku, Vercel, or AWS)?
- What is your experience with CI/CD pipelines?
- How do you use Git in team projects?
- How do you manage environment variables securely?

Testing

- What tools do you use for testing backend (e.g., Jest, Mocha)?
- How do you test React components?
- How do you handle unit vs integration testing?

System Design / Architecture

- How would you design a scalable MERN-based e-commerce application?
- How do you structure your folders in a MERN project?
- What are the best practices for API design in a full-stack app?
- How do you handle large file uploads/downloads in MERN?

Scenario-Based/Behavioral

- Describe a challenging bug you faced and how you fixed it.
- How do you approach a project from start to finish?
- Have you ever optimized a slow query or API? Explain how.
- How do you handle conflicts in a team setting?